Received: 8 June 2022 | Revised: 28 July 2022 | Accepted: 29 July 2022

DOI: 10.1111/jam.15754

Journal of Applied Microbiology sfam

ORIGINAL ARTICLE

Identification of NS5B resistance against SOFOSBUVIR in hepatitis C virus genotype 3a, naive and treated patients

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Abstract

Aims: Pakistan has the second highest prevalence of HCV with genotype 3a (GT-3a) being the most frequently circulating genotype. Currently, resistance-associated substitutions (RASs) are a major challenge in HCV treatment with direct-acting antivirals (DAAs). Sofosbuvir (SOF) is an FDA-approved NS5B nucleotide inhibitor. The aim of this study was to identify these RASs in the NS5B gene in naive and treated Pakistani HCV 3a isolates against SOF.

Methods and Results: Blood samples were collected from anti-HCV-positive patients, followed by HCV RNA isolation and real-time PCR quantification. HCV-positive patients were processed for HCV RNA genotyping, patients with genotype 3a were processed for NS5B gene amplification and sequencing. GT-3a was the most prevalent genotype (62.2%). S282T was identified in 2 (8.7%) patients, C316Y/G/R in 3 (13%), V321A and L320P in 1 (4.3%) each in SOF/RBV-resistant patients. Variants of S282 were detected in 3 (13%) of SOF/RBV-treated patients. While INF/RBV-associated mutations were also analysed, D244N, A333R and A334E were identified in 2 (9.5%), 3 (14.2%) and 7 (33.3%) in treatment-naive and 15 (65.2%), 7 (30.4%) and 5 (21.7%) treated patients, respectively. Q309R was observed only in one treatment-experienced patients. Some substitutions were present at higher frequency in both groups like N307G, K304R, A272D and R345H, considered that they do not have any role in sofosbuvir resistance.

Conclusion: It was concluded that sofosbuvir RASs are present in Pakistani HCV GT-3a isolates, and they should be monitored carefully, especially in treatment-experienced patients, for further selection of treatment regimens.

Significance and Impact of Study: HCV RASs have been studied very well across the world but there is scarcity of data regarding this topic in Pakistani population, this study provides data regarding the prevalence of these RASs in Pakistani HCV isolates emphasizing the fact that these RASs must be carefully monitored before starting HCV treatment, especially in treatment failure patients.

KEYWORDS

direct-acting antivirals, gene amplification, genotyping, hepatitis c virus, resistance-associated substitutions, sequence analysis

Saima Younas and Aleena Sumrin are contributed equally to this work.

INTRODUCTION

Hepatitis C virus (HCV) infects approximately 200 million people worldwide with 3–4 million new cases each year and 350,000 annual deaths due to HCV-related liver complications (Haqqi et al., 2019). Pakistan has the second highest prevalence of HCV with one of every 20 Pakistani infected with HCV (Mahmud et al., 2019). HCV is categorized into seven different genotypes (GT) and 67 subtypes. Genotype 1 (GT-1) and genotype 3 (GT-3) are worldwide in distribution and account for 46% and 30% of all HCV infections (Messina et al., 2015). European countries have a high prevalence of GT1, followed by GT-2 and GT-3, 75% of GT3 cases have been reported from South Asia, while East Asia is endemic with GT-2 and GT-6. GT-5 is very rare and has been reported from Eastern and Southern Africa (Haqqi et al., 2019). In Pakistan, GT-3 has been found to be the most prevalent genotype with a predominance of subtype 3a infecting approximately 61.3% of Pakistanis (Umer & Iqbal, 2016).

Hepatitis C virus is a small RNA virus of the Flaviviridae family, HCV genome consists of 9800 nucleotides that encode structural proteins C, E1 and E2 and seven non-structural proteins (Gededzha et al., 2017). NS5B is HCV RNA-dependent RNA polymerase synthesizing a new RNA chain from the RNA template (Ahmed et al., 2021). In 2011, direct-acting antivirals (DAAs) dramatically changed HCV treatment and made HCV cure possible with improved quality of life. DAAs are molecules specifically designed to target viral non-structural proteins leading to inhibition of protein and interruption in viral replication and infection (Zeng et al., 2020). Four different categories of DAA are available on basis of their target protein in HCV replication. (1) NS3/4A protease inhibitors that act by binding to active site of NS3/4A protease, (2) NS5A inhibitors that inhibit HCV replication by interaction with NS5A Domain-I, (3) NS5B nucleotide inhibitors become incorporated into the growing chain of RNA leading to chain termination and ultimately inhibition of viral replication and (4) NS5B non-nucleotide inhibitors inhibit polymerase activity by allosteric mechanism (Gerber et al., 2013; Pawlotsky, 2013).

Hepatitis C virus has a high replication rate and has RNA polymerase deficient in proofreading, leading to the introduction of mutations that result in resistance to DAAs therapy and affect response rates to these DAAs. Resistance-associated substitutions (RASs) are currently the major challenge in treating HCV patients with DAAs. These mutations can occur naturally in HCV treatment-naive patients and in patients resistant to DAAs treatment (Haqqi et al., 2019). These mutations have been linked to variable response rates to DAAs treatment among different HCV genotypes. HCV genotype 3a is the most frequent genotype in Pakistan and is considered less responsive to antiviral therapy (Haqqi et al., 2019).

Sofosbuvir (SOF) is an NS5B nucleotide inhibitor. RASs to SOF at positions S282T, L320F and C316Y in the NS5B gene have been reported to reduce virus susceptibility to anti-viral therapy (Jaspe et al.2012). The prevalence of these RASs has been broadly reported worldwide. In Pakistan, this issue has not been fully addressed like in other parts of the world. This study aimed to understand the underlying mechanism of drug resistance by identifying naturally occurring resistance mutations in the HCV NS5B gene in HCV genotype 3a treatment-naive and treatment-experienced patients from Lahore, Pakistan.

MATERIALS AND METHODS

Study population and sampling

This study was carried out at the Molecular diagnostics laboratory (MDL), Centre for Applied Molecular Biology (CAMB), the University of the Punjab Lahore, from January 2018 to December 2020. This study enrolled anti-HCV-positive patients from different hospitals and doctors referred to MDL CAMB. Following inclusion criteria were used for the selection of patient samples, (1) patients who were anti-HCV-positive and patients who were treatment-naive or had any kind of HCV treatment history (INF/RBV or DAA s treatment). Patients with co-infection with HBV or HIV were excluded from the study. Written approval was taken from each participant of the study. Each patient was interviewed for general information, and data were collected on a specifically designed questionnaire. This study with its protocols and patient consent form was approved by the Advance Studies and Research Board, University of the Punjab Lahore, Pakistan.

RNA Extraction and real-time PCR quantification

HCV RNA was isolated from plasma samples by using the QIAamp DSP Virus kit (Qiagen). Isolated RNA was amplified for HCV RNA quantification with an artus HCV PCR kit (Qiagen). Patient's samples were processed along with four HCV standards provided with kit, negative control and non-template control (NTC). Isolated RNA was stored at −20°C for next processing.

cDNA synthesis and HCV genotyping

First-strand cDNA was synthesized for patient samples that were found positive for HCV RNA. cDNA was synthesized for NSB amplification by using gene-specific antisense primer and 200 U/μl of Revert-Aid Reverse Transcriptase enzyme (ThermoFisher Scientific). cDNA for HCV genotyping was synthesized and genotyping was processed by Ohno et al. (1997) method.

HCV NS5B gene amplification by nested PCR

Specific primers were designed to amplify 421 bp fragment of the NS5B gene including codons for amino acids from 220 to 350. Amplification was carried out by nested PCR in two rounds, primers used for NS5B amplification are given in supplementary Table S1. These primers were utilized along with $MgCl_2$, 10x PCR buffer, 2.5 mM dNTPs and Taq DNA Polymerase (Thermo Fisher Scientific) to prepare PCR mix. Amplified products were pictured on 2% agarose gel along with a 100 bp DNA ladder (Thermofisher Scientific) and observed under UV light.

Purification of PCR product ad NS5B gene sequencing

The required band of 421 bp of amplified NS5B region was cut from gel and purified using Gene JET Gel Extraction Kit (Thermo Scientific). Sequencing of the eluted amplified PCR fragment was done by using gene-specific primers with big dye terminators (Big-Dye Deoxy Terminators, Applied Biosystems) on an automated genetic analyser.

Sequence analysis

Sequences obtained were first analysed by Bio-edit software (version 7.0). Amino acid sequence identification and resistance-associated substitutions (RASs) were selected utilizing geno2pheno [HCV] tool. Geno2Pheno [HCV] is a web-based tool for the identification of amino acid substitutions and is publicly available at http://hcv.geno2pheno.org. This publicly available tool is a web-based mutation detection algorithm for analysis of HCV sequence data w.r.t genotype and subtypes of HCV and for identification of clinically relevant amino acid substitutions against licensed DAAs.

Statistical analysis

Statistical analysis was done using SPSS (version 22), qualitative variables were expressed as percentage while quantitative variables were expressed as mean ± SD.

RESULTS

HCV quantification by real-time PCR

Initially, there were 324 anti-HCV positive samples collected for this study. From the detailed history of the patients, it was found that there were 231 patients who had SOF/RBV treatment for 24 weeks, and 93 patients were having HCV RNA PCR for the first time. From real-time PCR quantification of these 324 patients, it was found that 106 (32.7%) patients were positive for HCV RNA while 218 (66.3%) patients were negative for HCV RNA. The results of HCV RNA quantification have been shown in Figure 1a. Among 106 HCV RNA-positive patients 44 (42.4%) patients had undergone SOF/RBV treatment but they had relapsed HCV infection (could not achieve sustained virological response rate (SVR)), while the remaining 62 (58.6%) HCV RNA-positive patients were having their HCV PCR for the first time. Among 218 HCV RNA-negative patients 187 (85.7%) were treatment-experienced and 44 (14.3%) patients had no history of HCV treatment. These frequencies of HCV RNA-positive and -negative patients along with their treatment status are described in Figure 1b.

HCV genotyping by nested PCR

Samples found positive for HCV RNA were processed for HCV genotype determination. Genotyping results revealed that there were 66 (62.2%) samples that were of HCV GT-3a, 17 (16.2%) samples of GT-1a, 13 (12.2%) of GT-3a/3b and 10 (9.4%) were untypable. Frequencies of different genotypes are shown in Figure 2.

Amplification of the HCV-NS5B region by nested PCR

As HCV GT-3a was found to be the most common genotype in the study group, so this GT remained the focus of this study. 66 samples of genotype 3a were processed for HCV NS5B amplification and sequencing. 44 patients could be successfully amplified, sequenced and available for analysis. Based on treatment history these 44 sequences were divided into two groups (1) there were 21 patients who have not received any kind of HCV treatment and were grouped as treatment-naive and (2) 23 patients who had undergone SOF/RBV treatment but could not produce SVR and grouped as treatment-experienced. Epidemiological and viral features of these 44 patients are listed in Table 1.

Mean age was 41.7 ± 14.5, and 43.1 ± 14.7 in treatment-experienced and -naive groups, respectively. 11 males

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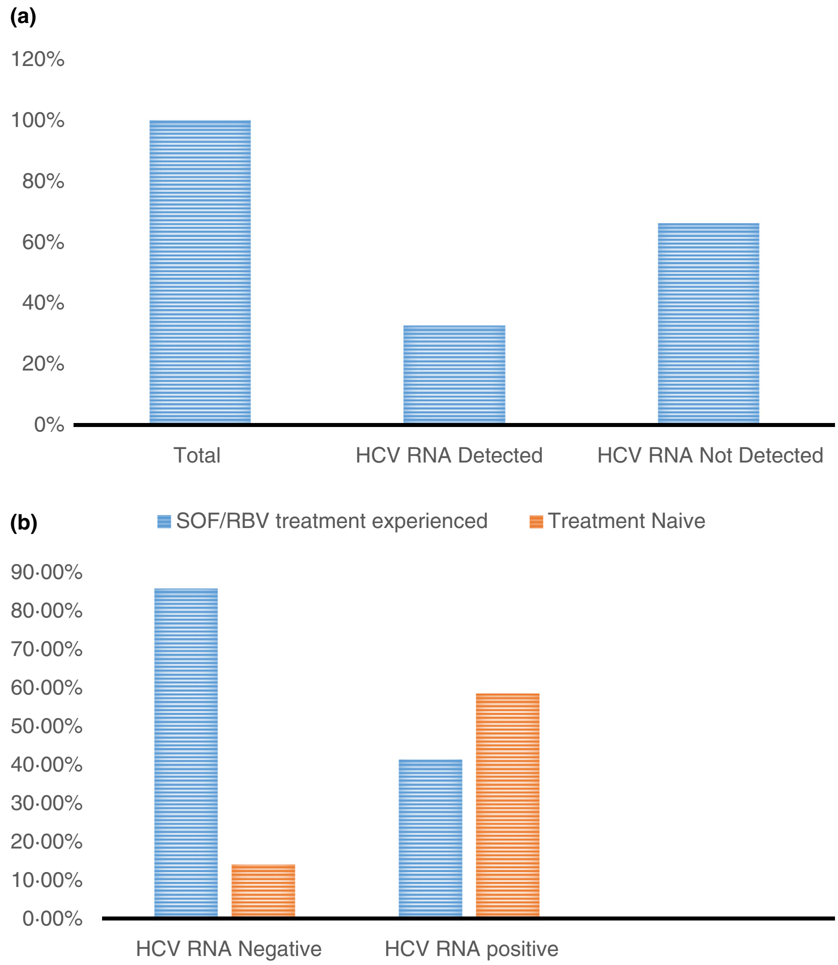

FIGURE 1 (a) HCV real-time PCR quantification, frequency of HCV RNA-positive and -negative samples. (b) Frequency of treatment-experienced and treatment-naive patients in HCV RNA-positive and HCV RNA-negative patients.

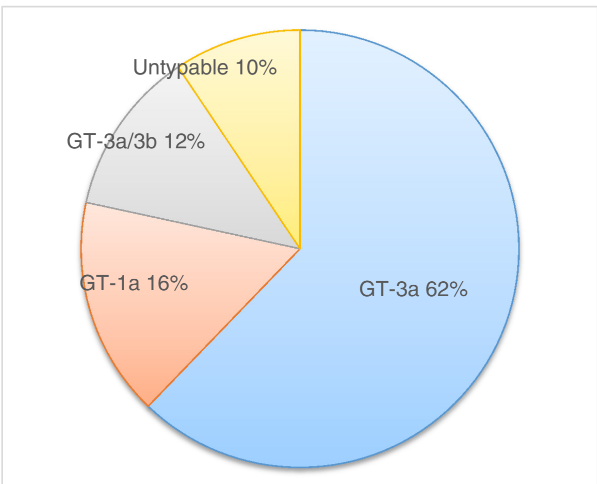

FIGURE 2 Distribution of different genotypes in the study group.

and 12 females were in the treatment-experienced group and 10 males and 11 females were in the treatment-naive group. Mean viral load was 5.48 ± 1.8, and 5.80 ± 1.7 in treated and non-treated patients, respectively. In Pakistan, sofosbuvir (SOF) is the easily available polymerase inhibitor that is mainly prescribed by the physician for HCV treatment, positions in the NS5B region having resistant-associated variants to SOF were mainly examined. Other positions that are associated with resistance to dasabuvir were not included in the study.

Analysis of sofosbuvir-associated resistance variants in treatment-experienced group

Amino acid mutation at position 282 (S282T) is considered to contribute to extreme level of resistance to SOF, 2 (8.7%) sequences of the treatment-experienced group were found to have this mutation. In the remaining sequences of this group this primary mutation associated with resistance to SOF was not detected, however other S282 variants were observed like S282I, S282F and S282R in one sequence each (4.3%) collectively in 13% sequences. One patient carrying S282T substitution was 27 years male having viral load of 803,069 IU/ml, while the second was 45 years female with HCV viral load of 348,015 IU/ml. The patient with with S282I variant was 35 years female with viral load of 9043 IU/ml, one with S282F was 47 years male with 265,022 IU/ml viral load

TABLE 1 Epidemiological and viral features of treatment-naive and -treated patients

Parameters	Treatment-experienced group, $N = 23$	Treatment-naive group, $N = 21$
Age + (years)	41.7 + 14.5	43.1 + 14.7
Gender male/female	11/12	10/11
Viral load (IU/ml)	5.48 + 1.8	5.80 + 1.7
HCV genotype	3a	3a

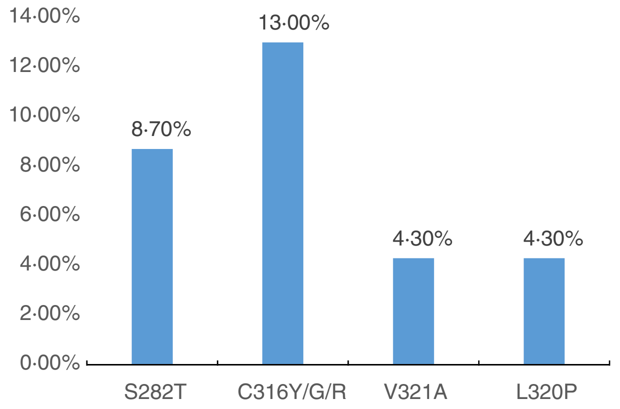

FIGURE 3 Prevalence of SOF RASs in the treatment-experienced group.

and the one carrying S282R variant was 66 years male with 3,078,692 IU/ml viral load.

C316YG/R polymorphism was present in 13% of sequences, L320P and V321A substitutions were identified in 4.3% of sequences each. Sofosbuvir-associated mutations with their frequency are shown in FIGURE 3. In addition to these RASs some substitutions at scored positions were also observed, these all substitutions with the amino acid in reference strain and mutated amino acid are described in Table 2.

Analysis of amino acid substitutions in treatment-naive group

Substitutions playing role in resistance to SOF were not identified in yhe treatment-naive patient group. However, some other substitutions were observed in this group that were found to be present in the treatment-naive as well in treatment-experienced group. A full list of substitutions identified in both groups has been provided in supplementary Table S2. Some of the important substitutions with higher frequency are recorded in Table 3.

N307G and K304R are the substitutions that were found at higher frequency in both groups. D244N has been identified at the frequency of 9.5% and 65.2% in treatment-naive and treatment-experienced groups, respectively. A333R was detected in 14.2% and 30.4% of treatment-naive and -experienced groups, while the frequency of A334E was 33.3% and 21.7% in treatment-naive and -experienced patients. A272D and R345H are amino acid substitutions that were identified at higher frequency in both groups. As these substitutions are identified in both the groups, it was considered that they present naturally at baseline level. The prevalence of these naturally occurring substitutions in both groups has been depicted in Figure 4. Q309R is an RBV-associated resistance mutation that was also identified in 4.3% sequences of this group.

DISCUSSION

The introduction of DAAs in the HCV treatment has significantly increased response rates (up to 98%) and greatly reduced treatment duration (Asselah et al., 2018). However, the extreme mutations and high replication rates of HCV together with the immune system pressure, lead to a remarkable genetic variability that can compromise the high response rates to DAAs due to the preexistence of RASs (Esposito et al., 2016; Wyles, 2017). Due to the scarcity of data available on these RASs in Pakistan particularly in GT-3a, we evaluated these RASs in treatment-naive and treatment-experienced HCV 3a isolates from Lahore Pakistan.

This study reports 66.3% negative samples and 32.7% positive for HCV RNA. From the history of these HCV negative samples, that was already taken at the time of sample collection on prescribed questionnaire it was observed that 85.7% of these patients had a history of SOF/RBV treatment, while the remaining 14.3% of patients with negative HCV RNA were treatment-naive and first time going through real-time PCR quantification due to positive result of anti-HCV. Serum anti-HCV and HCV RNA are initial indicators for HCV infection, anti-HCV is sensitive and easy to perform while HCV RNA detection is the monetary standard for clinical diagnosis and antiviral therapy. There can be different reasons for negative HCV RNA in anti-HCV-positive patients. HCV antibody detection is considered more specific and sensitive method for the identification of HCV infection, but frequent false-positive results have also been reported with anti-HCV

TABLE 2 Sofosbuvir-associated resistance substitutions in NS5B gene of SOF/RBV treatment-experienced group

Amino acid in reference strain	Position in NS5B protein	Mutation/substitution	RAS name	% age
S	282	T	S282T	8.7%
C	316	Y/G/R	C316Y	13%
V	321	A	V321A	4.3%
L	320	P	L320P	4.3%
Substitutions at scored positions				
S	282	R/I/F	S282R/I/F	13%
V	321	F	V321F	4.3%

TABLE 3 Amino acid substitutions in NS5B gene identified in treatment-naive and treatment-experienced group

Amino acid in reference strain	Position in NS5B protein	Mutation/substitution	Substitution name	Treatment-naive group no (% age)	Treatment-experienced group no (% age)
N	307	G	N307G	20 (95.2)	22 (95.6)
K	304	R	K304R	16 (76.1)	19 (82.6)
D	244	N	D244N	2 (9.5)	15 (65.2)
A	272	D	A272D	7 (33.3)	6 (26.0)
L	308	F	L308F	2 (9.5)	6 (8.6)
Q	309	R	Q309R	—	1 (4.3)
D	330	E	D330E	4 (19.0)	4 (17.3)
A	333	R	A333R	3 (14.2)	7 (30.4)
A	334	E	A334E	7 (33.3)	5 (21.7)
R	345	H	R345H	5 (23.8)	10 (43.4)

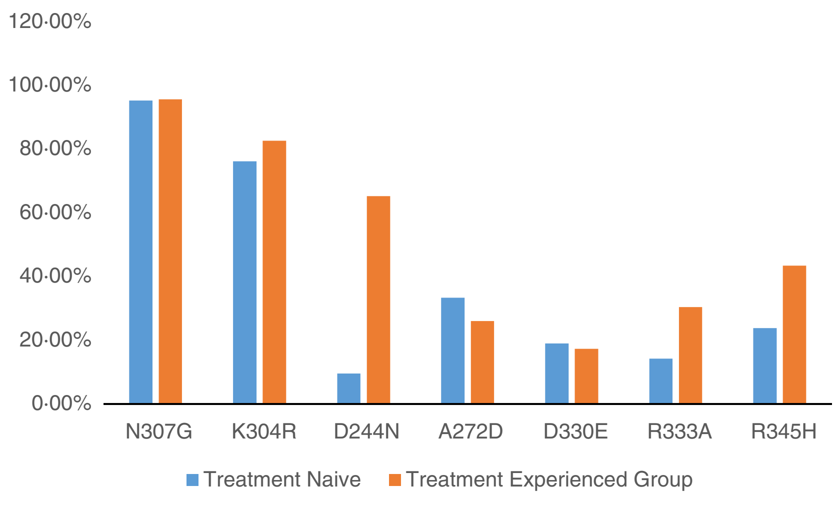

FIGURE 4 Naturally occurring amino acid substitutions of NS5B gene of HCV in HCV treatment-naive and treatment-experienced groups.

assays, or it may be the virus has been cleared naturally by the immune system of the anti-HCV-positive patient (Mohan et al., 1999).

In the present study, GT-3a was the most commonly detected genotype of HCV. Previous studies have also reported GT-3a as the most prevalent genotype in Pakistan. The results of our study are in accordance with Ahmad et al study who reported in their study 59.9% of cases of GT-3a and 16.5% of GT-1a (Ahmad et al., 2010). Other studies have also reported that genotype 3a occurs most frequently in Pakistan (Afridi et al., 2014). This study reports mutation in the NS5B gene of HCV encompassing codons 220–350, including all important amino acid positions that are considered to involve in resistance to

SOF. For the best treatment of HCV, especially in non-respondents, detection of baseline prevalence of HCV RASs such as S282T is important.

S282T is considered to provide intense grade resistance to SOF in vitro. In the present study, S282T mutation was detected in 8.7% sequences from SOF/RBV treatment-experienced group, while it has not been identified in any sequence from the treatment-naive group. S282T in the NS5B region of HCV is the primary mutation chosen by SOF in in vitro studies. It has been indicated in an in vitro study that S282T substitution resulted in 9.5-fold resistance to SOF (Lam et al., 2012). Previously, there is only one study that described this S282T mutation in one patient with HCV (Franco et al., 2013; Gane et al., 2013). In a large-scale analysis of 1344 HCV isolates focusing on the NS5B gene, S282T was identified only in one isolate from each genotype 1a, 1b, 3 and 4, so the findings of the present study regarding S282T mutation are in accordance with the previous studies reporting low prevalence of S282T mutation (Franco et al., 2013).

Previously C316Y and L320F, V321A substitutions are formerly documented as resistance mutations to SOF, in the present study, these RASs were identified in 4.3% sequence each. The present study reports C316Y/R/G polymorphism in 13% of treatment-experienced patients. Mutation at C316 confers a low level of resistance to sofosbuvir and this substitution has been reported at low frequencies in previous studies (Di Maio et al., 2014; Lontok et al., 2015). A Venezuelan study reports that C316N polymorphism was present in 18% of Venezuelan HCV isolates, while this mutation has been reported to be present at a higher frequency in Asian isolates (Jaspe et al., 2012)˙

V321A is considered SOF treatment-emergent substitution and analysis of this variant in NS5B modelled crystal structure indicated that this variant is located close to SOF binding site and could possibly affect the anti-HCV activity of SOF. Svarovskaia et al., 2014 have reported in their study emergence of V321A variants in 5% of HCV-GT3a patients. L320F substitution was not observed in our study group however L320P variant was observed at this position and the role of this variant in the development of SOF resistance is not clear, more studies are needed to identify L320F substitution and its variants at this position and their role in SOF resistance in Pakistani HCV GT-3a isolates (Svarovskaia et al. (2014)).

In addition to these RASs, some substitutions at these scored positions were also identified like S282R/I/F and V321F. As it is clear that S282T is a potent SOF-associated RASs but variants at this position also lead to the development of the same level of resistance that need more attention and research. The present study reports 13% S282 variants that is in agreement with the study of Hezhao et al. (2015) also reported S282 variants at different frequencies and evaluated their role in resistance to SOF. They reported in their study that S282R or S282G variants can affect SOF binding in a similar way as S282T leading to SOF resistance (Hezhao et al., 2015). Donaldson et al. (2015) also reported S282R in a patient who failed a sofosbuvir treatment. Viruses with this substitution, are rare, but may lessen the effectiveness of SOF and effect future treatment options, further studies are required to determine the direct role of these variants in viral fitness and drug resistance (Donaldson et al., 2015)˙

In addition to SOF-associated RASs, some INF/RBV-associated substitutions were also observed. D244N amino acid substitution has been previously reported to be associated with resistance to RBV. D244 amino acid is located at flexible loop connecting two helices of NS5B protein and can interfere with protein–protein interaction of NS5B protein. This D244N substitution has been extensively reported in HCV GT-3a patients, is considered to be associated with RBV resistance and also had been reported in treatment-naive patients (Asahina et al., 2005; Castilho et al., 2011). The findings of the present study are in accordance with already reported studies as D244N has been recorded in 9.5% treatment-naive and 65.2% treatment-experienced patients.

Three other substitutions were observed in both groups Q309R, A333R and A334E have been previously reported to be involved in RBV resistance (Jaspe et al., 2012). Q309 and A333 amino acids are located at the surface of the NS5B catalytic domain (Asahina et al., 2005; Castilho et al., 2011). In the present study, Q309R was detected in 4.3% sequences of treatment-experienced but it was not identified in any of treatment-naive patients. Previous studies have reported this Q309R to be highly prevalent in HCV GT 1a than in HCV GT-3a (Asahina et al., 2005). A333R was detected in 14.2% and 30.4% of treatment-naive and -experienced groups, respectively, literature search shows that this RBV-associated resistance mutation and is also highly prevalent in all HCV GT (Hamano et al., 2005)˙

Some amino acid substitutions have been identified in the present study that are present in both treatments-naive and treatment-experienced groups. N307G and K304R are substitutions that were present in higher frequency in both groups. L308F, D330E, R345H and A272D were also identified at different frequencies in both groups, indicating that these mutations are present at baseline as they are detected in treatment-naive as well as treatment-experienced patients. Naturally occurring mutations in NS5B region have been reported in some studies, like in an Argentine study reports a 6.3% prevalence of RASs in NS5B region of HCV (Martınez et al., 2019). Prevalence of RASs was reported to be 20.9% in another study, in which 43 HCV strains underwent extra analysis of the HCV NS5B by next-generation sequencing (Tameshkel et al., 2020).

Mutations at codons K304, N307 and L308F had been reported in 5 strains of HCV out of 10 studied strains in another study (Asadpour et al., 2019). Also, more than one substitution at the same amino acid positions was noticed among present study HCV patients. It is clear from this study that some newfound baseline NS5B substitutions may not affect treatment outcomes as they are enriched not only in treatment-experienced but also in treatment-naive patients.

This study reports S282T, C316Y/G/R and V321A RASs in the NS5B gene of HCV GT-3a in Pakistani HCV isolates. INF/RBV-associated substitutions like D244N, A333R and A334E, with some other substitutions like K304R, N307G, A272D and R345H that are present at higher frequency in treatment-naive HCV patients as well as in treatment-experienced patients indicating that they are present naturally in Pakistani HCV GT 3a isolates. As they are present at baseline so it can be concluded that they do not play any role in SOF resistance. In conclusion, sofosbuvir RASs S282T has been detected in treatment-experienced patients and V321A and C316Y/G/R have also been identified in treated patients. It indicates increased resistance to sofosbuvir leading to failure of this treatment regimen. Identifying these RASs is crucial for the selection of re-treatment strategies, further large-scale studies with deep sequencing technologies should be conducted for more insights of the RASs, and to access treatment failure rate and re-treatment success rate.

ACKNOWLEDGEMENT

The authors would like to acknowledge CAMB, University of the Punjab and Higher education commission (HEC) of Pakistan for facilitating this research work.

CONFLICT OF INTEREST

The authors have no conflict of interest.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

How to cite this article: Younas, S., Sumrin, A., Hussain, N. & Bilal, M. (2022) Identification of NS5B resistance against SOFOSBUVIR in hepatitis C virus genotype 3a, naive and treated patients. *Journal of Applied Microbiology*, 133, 2826–2834. Available from: https://doi.org/10.1111/jam.15754